# Detailed Analysis of Sensors and Actuators

Robotics Engineering Fundamentals | Systems Integration

Technical Presentation

# Vision Sensors

## Environmental Perception

A vision sensor is a device that collects visual information from its surroundings and processes it to perform complex activities.

- **Object Detection:** Identifying entities in space.
- **Recognition:** Classifying objects based on features.
- **Tracking:** Following moving targets.

Includes Camera sensors for image capture and Infrared (IR) sensors for object detection and tracking.



# Tactile and Temperature Sensors

## Tactile Sensors

Collects data from physical interactions with the environment. Designed after the biological sense of cutaneous touch.

- **Force Sensors:** Measures load or force applied.
- **Touch Sensors:** Mainly used in touch screens and robotic pads.

## Temperature Sensors

Detects changes in temperature through a resistor or thermocouple.

- Used in industrial applications.
- Maintains set temperature limits in robotic internal enclosures.

# Range and Proximity: Ultrasonic

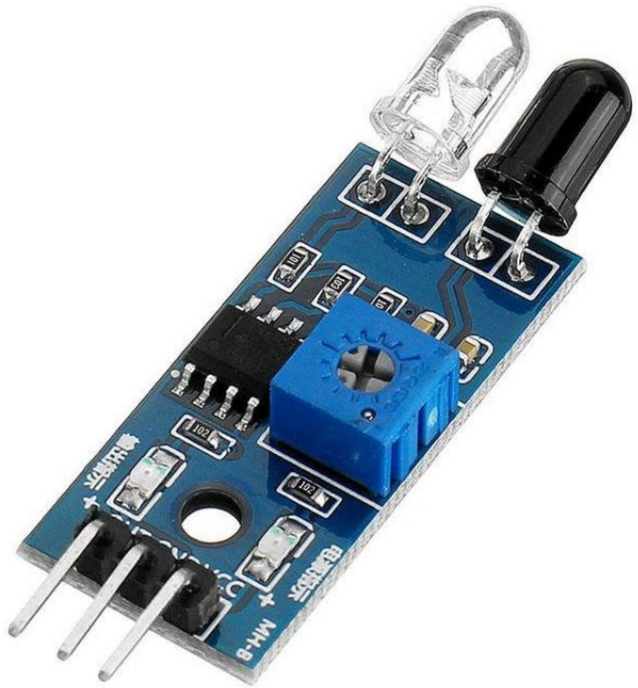
## Ultrasonic Sensors

These sensors use sound waves to detect objects and measure the distance to a target object.

- Non-contact distance measurement.
- Commonly used in parking sensors.
- Provides distance by measuring time-of-flight of sound waves.



# Range and Proximity: IR Sensors



## Proximity Sensors

Detects the presence of an object when it enters the sensor's field using light (IR), sound, or magnetic fields.

- Used in mobile phones to detect ear proximity.
- Essential for home appliances and obstacle avoidance.
- Works without physical contact.

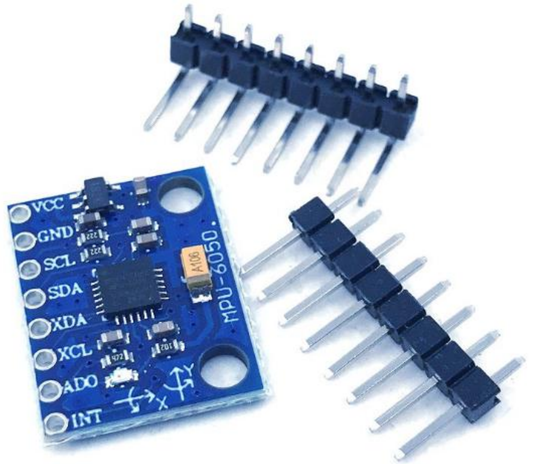
# Motion Sensors: Accelerometer & Gyro

## Accelerometer

Measures acceleration and tilt. Essential for auto-rotation in smartphones and track steps in wearables.

## Gyroscope

Measures angular velocity. Vital for drones, virtual reality systems, and stability in gaming controllers.




digipurk.ee

# Navigation Sensors




## Global Positioning System (GPS)

Provides information on position, velocity, and direction for travel by ocean, road, or air.

- Used in autonomous vehicles and drones.
- Connects to a network of satellites for geolocation.
- Critical for long-range robot navigation.

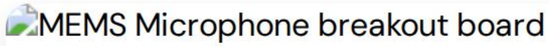
# Speech and Audio Processing

## Microphone Array

Collects audio signals for voice commands and communication. Essential for voice-controlled robots.

## Voice Activity Detector

Detects the presence of speech in audio signals. Helps in filtering background noise for voice assistants.

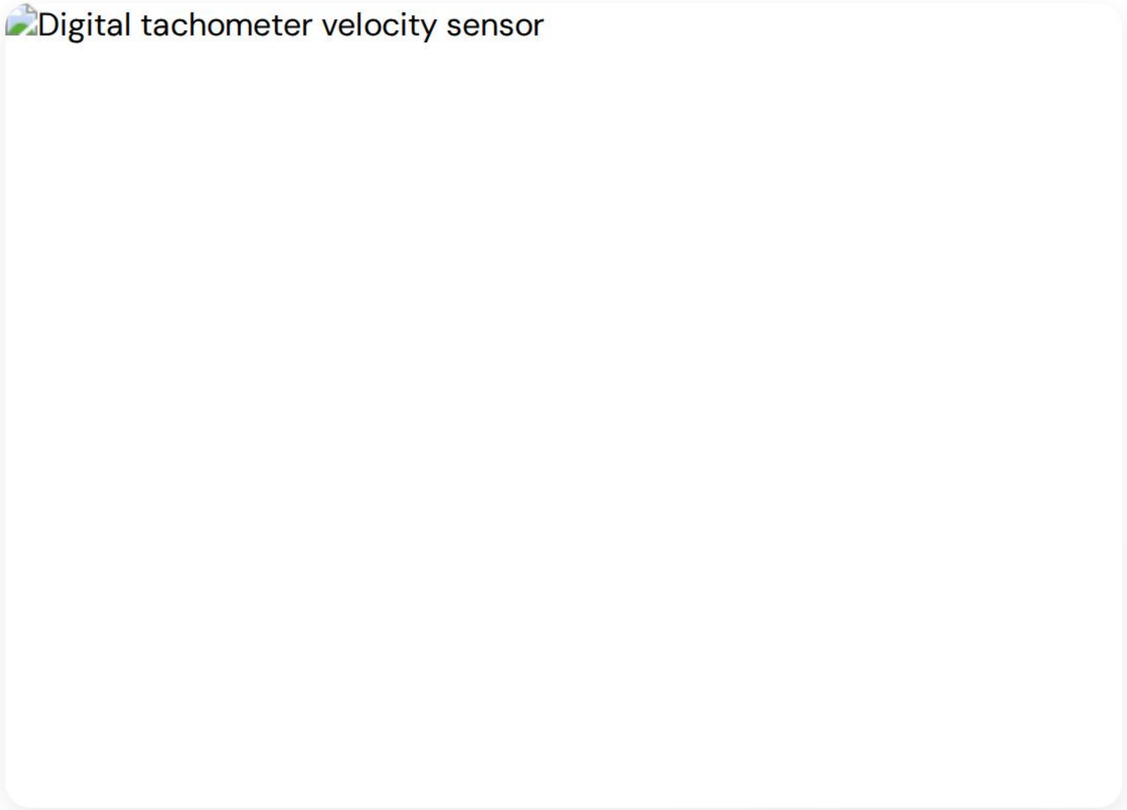


# Velocity and Acceleration Sensing

## Velocity Sensors

Measures the rate of change of location over time. Monitors object speed or rotation velocity.

- **Tachometers:** Monitors rotation of wheels.
- **Doppler Radar:** Measures speed via frequency shifts.



# Core Functions of Sensors

## Transform

Converts physical parameters like temperature or pressure into electrical impulses.

## Quantify

Precisely measures the value of the physical parameter perceived.

## Detect

Identifies changes or occurrences in the environment that can be utilized.

## Monitor

Constantly reports on changes in the measured parameter over time.

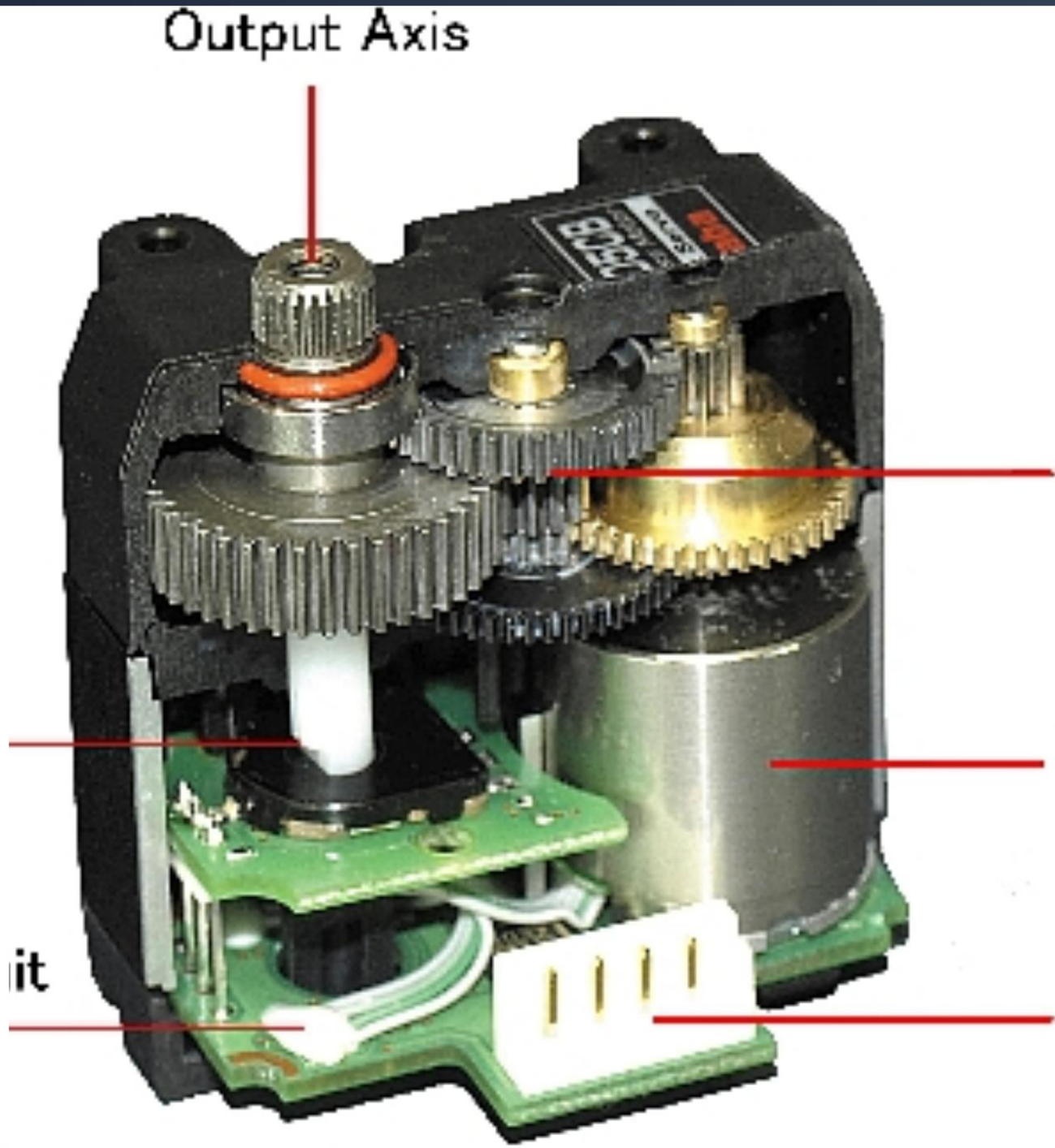
## Feedback

Provides data to control systems to maintain settings or adjust to changes.

# Actuators: From Signal to Motion

Actuators are devices that convert electrical, hydraulic, or pneumatic impulses into **physical motion or force**.

> *They are the necessary components that play an important role in achieving movement and functionality in robotics and aerospace.*



# Questions?

Thank you for attending this detailed overview of robot sensing.

Source: Components of Robots as a System | Robotics Unit 4

# Image Sources



https://image.made-in-china.com/202f0j00OEhBJmFnGHrI/Hikrobot-Mv-Sc3013xc-12m-Wbn-1-3MP-Industrial-Machine-Vision-Sensor-with-60fps-Global-Shutter-and-Profinet-Camera.webp

Source: xy-pyrometer.en.made-in-china.com

---



https://www.sparkfun.com/media/catalog/product/cache/f8158826193ba5faa8b862a9bd1eb9e9/1/5/15569-Ultrasonic_Distance_Sensor_-_HC-SR04-01a.jpg

Source: www.sparkfun.com

---



https://projects.arduinocontent.cc/cover-images/2a9dd489-3a70-4928-bf5a-a289106d2f3a.blob

Source: projecthub.arduino.cc

---



https://digipurk.ee/wp-content/uploads/2026/04/336-01.png

Source: digipurk.ee

---



https://www.voltaat.com/cdn/shop/files/voltaat-clearance-sale-ublox-neo-m8n-gps-positioning-module-gps-navigation-module-1243139879_1024x1024.jpg?v=1780915512

Source: www.voltaat.com

---

Thumbnail for uk.robotshop.com

http://uk.robotshop.com/cdn/shop/files/knowles-elecrow-digital-mems-microphone-breakout-board-sph0641lu4h-1-img1_1200x1200.webp?v=1720488800

Source: uk.robotshop.com

# Image Sources



https://image.made-in-china.com/365f3j00ktaWBMlcACbn/Digital-Contact-Tachometer-Rotate-Speed-Linear-Velocity-Motor-Frequency-Meter.webp

Source: dataloggeryuwen.en.made-in-china.com

---



https://d21xsz2za2b5mz.cloudfront.net/2916/2916/9179/RS405CB_Cut_04mini.jpeg

Source: www.futaba.co.jp